Lecture 1

CME 296: Diffusion & Large Vision Models

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Stanford University

Teaching staff

Afshine and **Shervine**

Teaching staff

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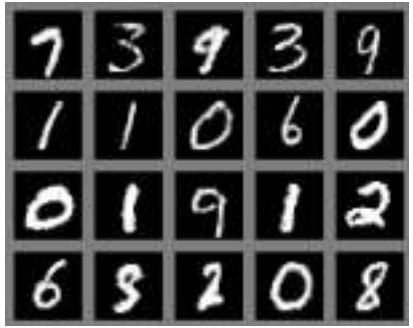
Centrale Paris ('16), MIT ('17)
Uber, Google, Netflix

Shervine
Centrale Paris ('16), Stanford ('19)
Uber, Google, Netflix

Motivation behind CME 296



Images from Generative Adversarial Networks, Goodfellow et al., 2014.

Motivation behind CME 296

Images generated by ChatGPT, March 2026.

Motivation behind CME 296

?

CME 296 overview

Goals.

1. Understand **paradigm** behind how we generate images
2. Learn **how** image generation models are **trained** & **used**

CME 296 overview

Goals.

1. Understand **paradigm** behind how we generate images
2. Learn **how** image generation models are **trained** & **used**

Audience. Interested in state-of-the-art image generation models

1. Career goal
2. Personal project
3. Curiosity

Prerequisites

- **Linear algebra**

  Vectors, matrices, gradient, divergence

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  Bayes' rule, conditional/marginal probability, expectation, covariance, Gaussian distribution

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  Ordinary/stochastic differential equations, numerical solvers

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- **Probability theory**

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- **Differential equations**

  Ordinary/stochastic differential equations, numerical solvers

- **Basics of machine learning**

  Loss function, training, inference, neural network

Logistics

Date & time.

- Fridays from 3:30pm to 5:20pm
- Thornton 110

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- Fridays from 3:30pm to 5:20pm
- Thornton 110

Details about the class.

- 2 units, Letter or Credit/No credit
- Lectures are recorded
- 2 exams (example for the LLM class)
  - Midterm (50% grade), scheduled on Friday May 1st
  - Final exam (50% grade), scheduled on Monday June 8th

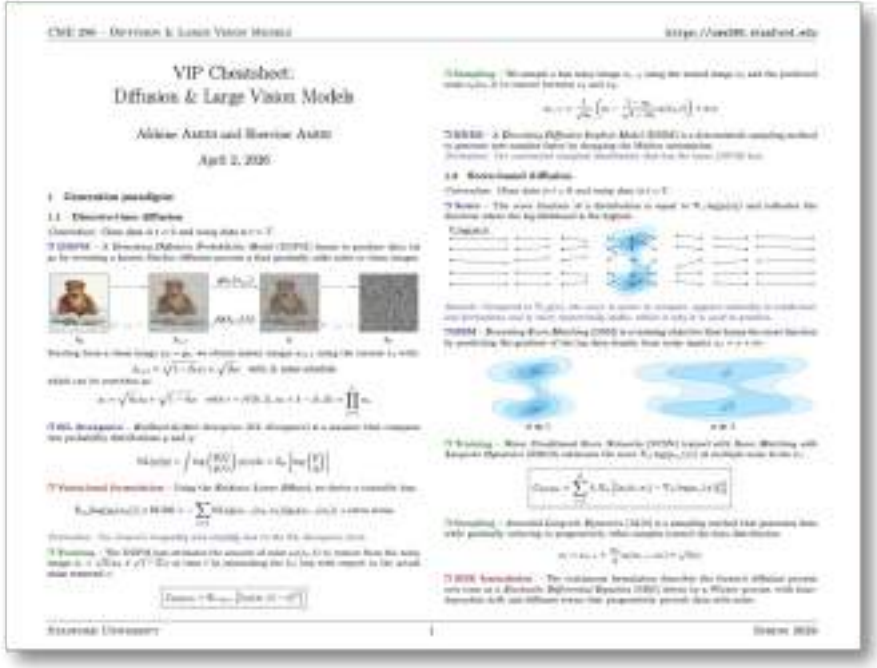
Material

Class website. cme296.stanford.edu

- Contains syllabus & logistics
- Slides and recordings will be posted there

Material

VIP Cheatsheet



Link to PDF: https://github.com/afshinea/stanford-cme-296-diffusion-large-vision-models

Class communications

Canvas.

- Announcements
- Class discussions via Ed

Any other general inquiries / questions, email us:

- cme296-spr2526-staff@lists.stanford.edu
- afshine@stanford.edu, shervine@stanford.edu

Example of a slide in this class

Explanation of a CME 296 concept

Example of a slide in this class

Explanation of a CME 296 concept

Source & suggested reading, if interested

Some paper

Important note on conventions

⚠️ Field is full of different notations / conventions ⚠️

$t \to [0, 1]$, $t \to [0, T]$

$\epsilon \to z, x_0, x_1, \ldots$

etc.

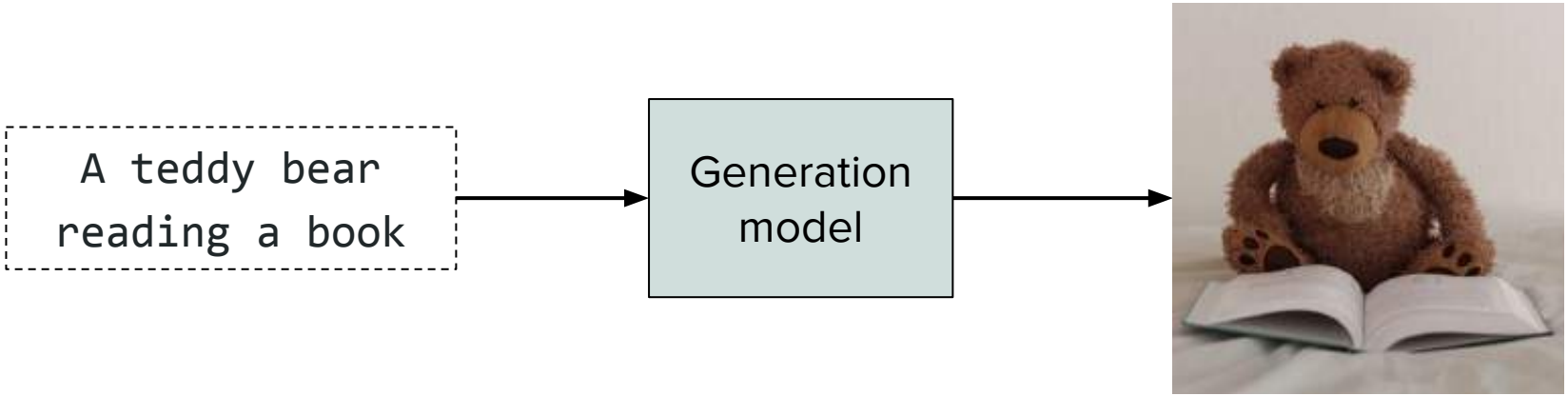
CME 296 will (try to):

1. Rely on the **most commonly-used** notations

2. **Stay consistent** throughout the class when that makes sense

3. **Call out** unintuitive change in notations

Outline of this class



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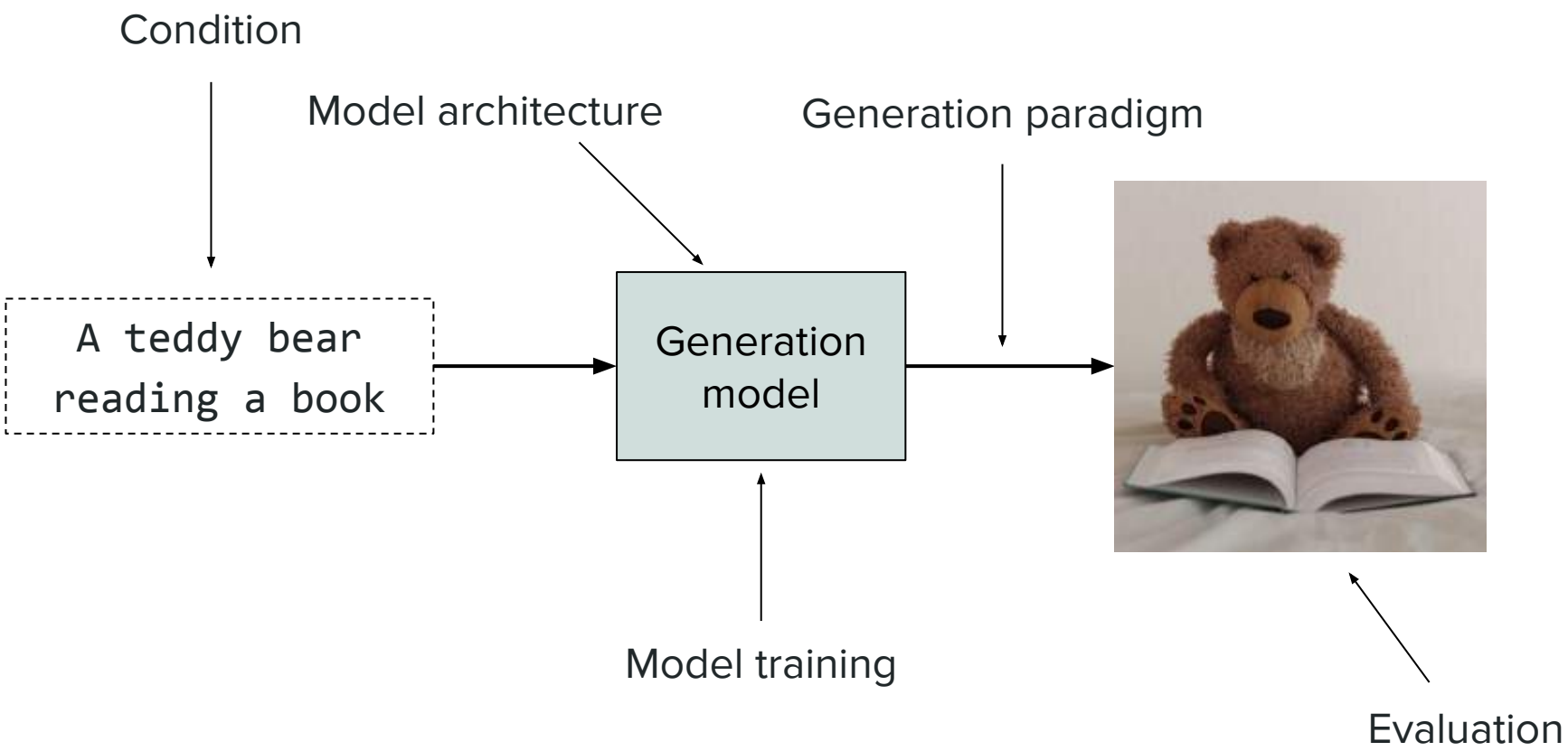

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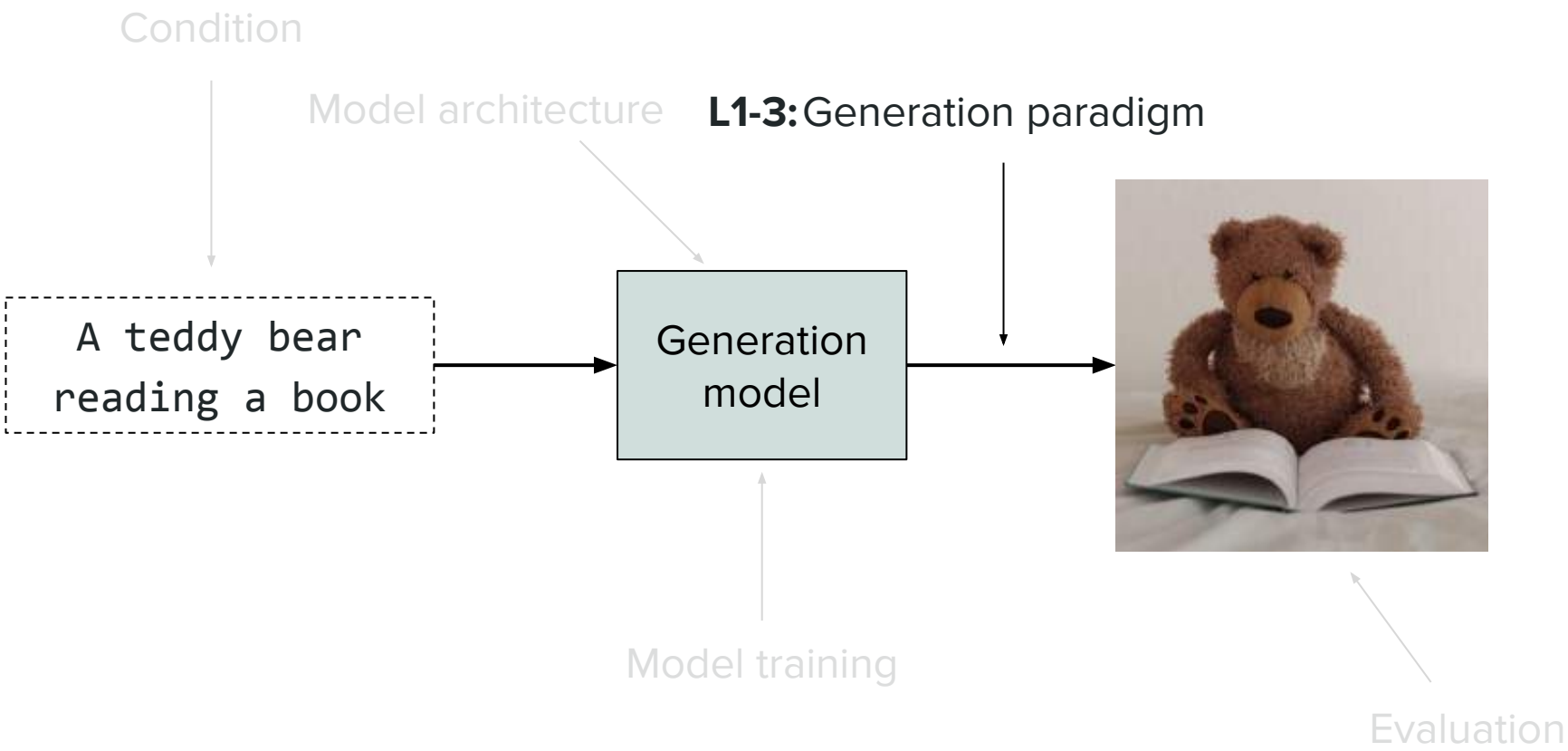

Outline of this class

L4: Condition

Model architecture

L1-3: Generation paradigm

A teddy bear reading a book

Generation model

Model training

Evaluation

Outline of this class

L4: Condition

L5: Model architecture

L1-3: Generation paradigm

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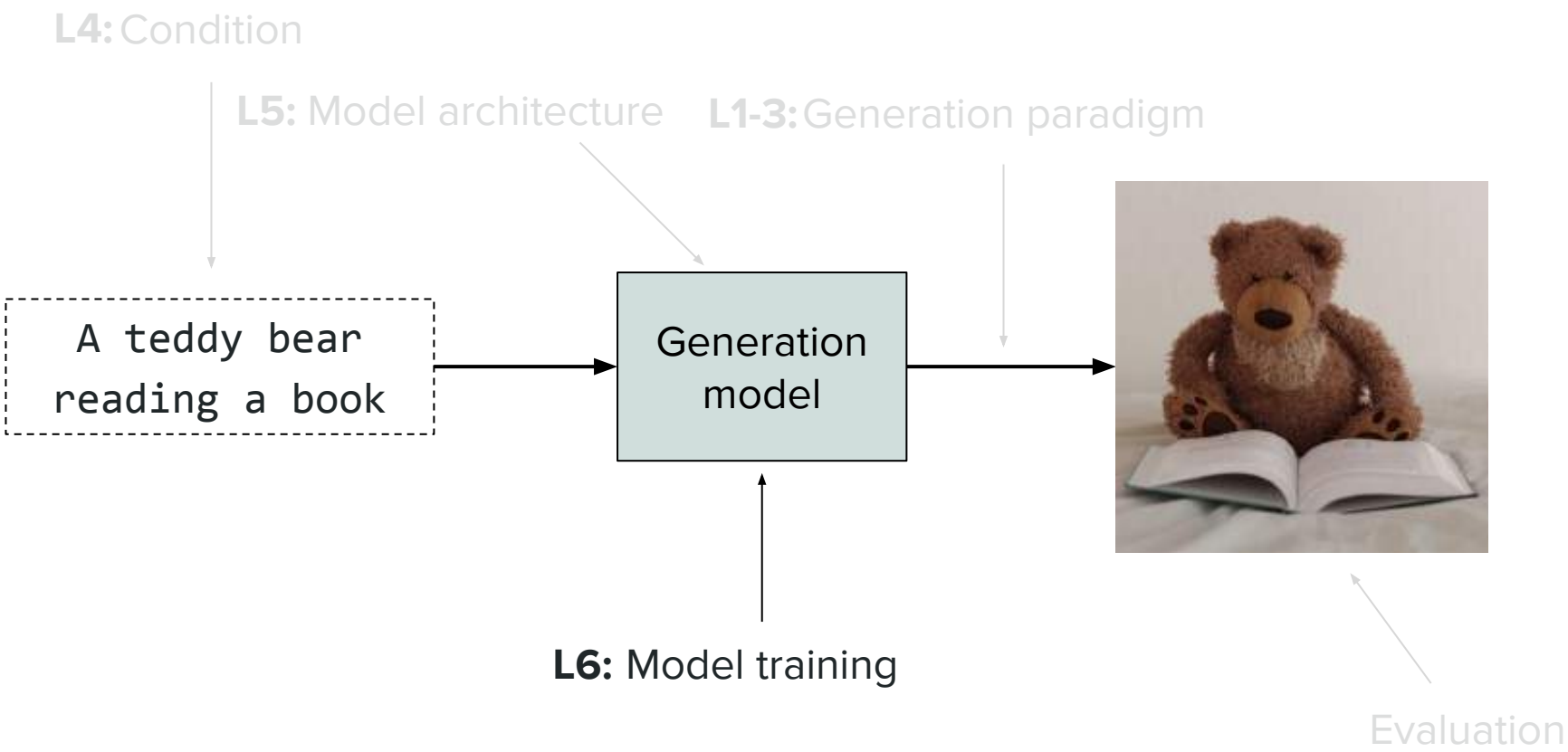
Outline of this class



Outline of this class

L4: Condition

L5: Model architecture

L1-3: Generation paradigm

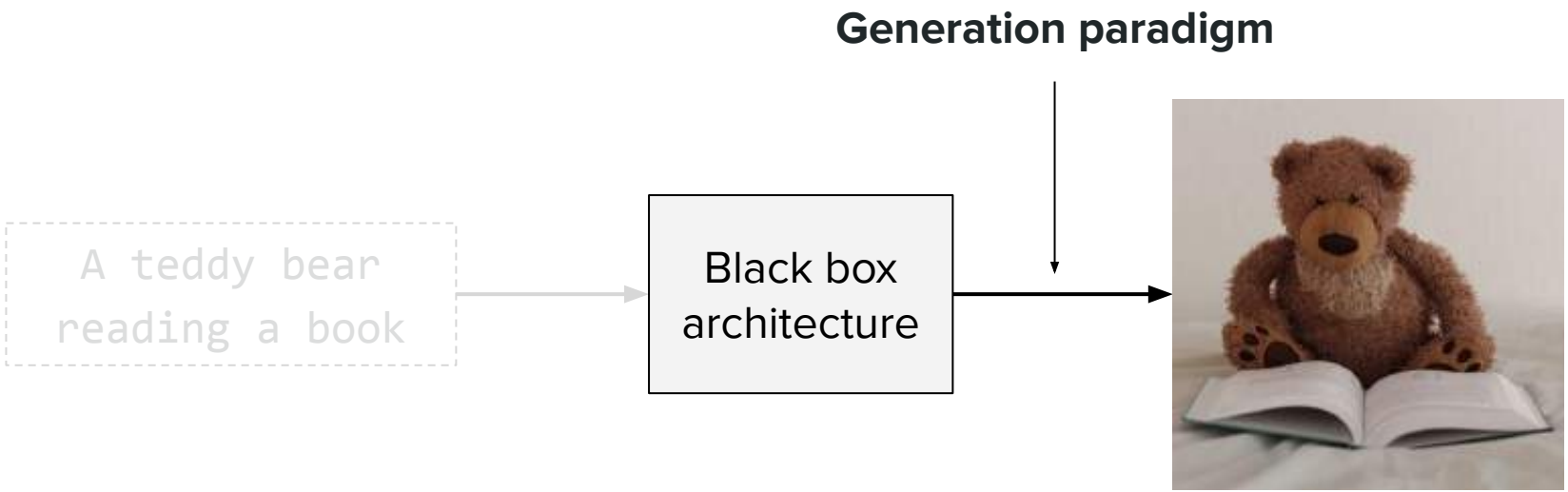
A teddy bear reading a book

Generation model

L6: Model training

L7: Evaluation

First three lectures: generation paradigms



Today's lecture: **Diffusion**

Diffusion & Large Vision Models

Motivation

Forward / backwards process

Variational formulation

Training

Inference

Faster sampling

Problem formulation

Suppose we are given a **sample** of observations from p_{data}

Sample

New generation

Objective. Generate new images from a distribution of images p_{data}

Problem formulation

Suppose we are given a **sample** of observations from p_{data}

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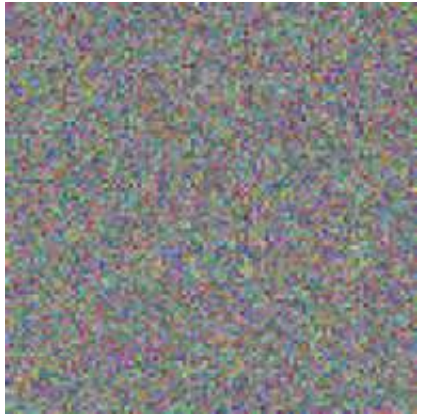
New generation

Objective. **Generate** new images from a distribution of images p_{data}

Unconditioned ← Focus of the 3 first lectures

Strategy: generate from noise



Why start from noise?

- Noise is **easy** to sample
- Introduces **randomness**
- Gaussian distribution has **nice properties**

Analogy: building a sculpture

The sculpture is already complete within the marble block, before I start my work. It is already there, I just have to chisel away the superfluous material.

–Michelangelo

Diffusion & Large Vision Models

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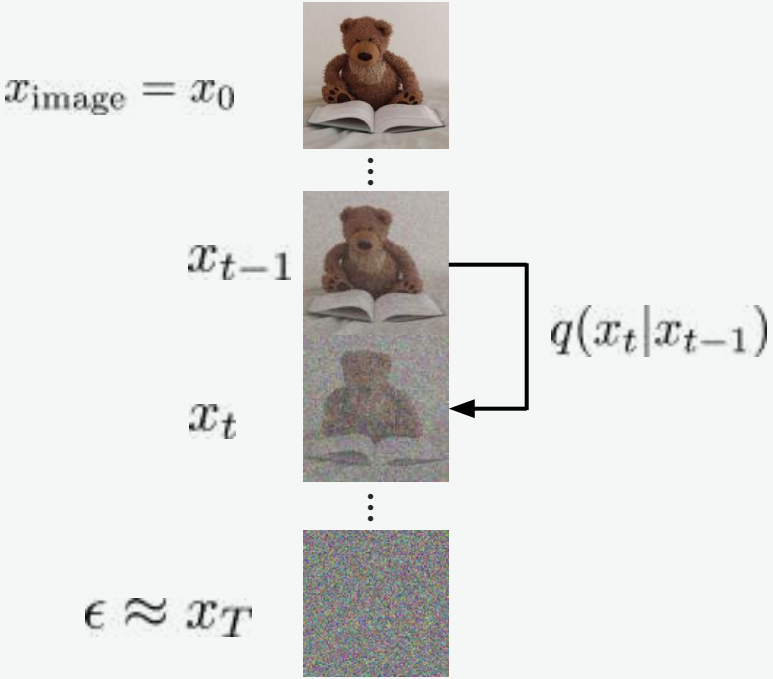
Intuition behind diffusion

1. Add **noise** (**forward process**)



Intuition behind diffusion

1. Add **noise** (**forward process**)

$x_{\text{image}} = x_0$

x_{t-1}

x_t

$\epsilon \approx x_T$

We know

$$q(x_t|x_{t-1})$$

Intuition behind diffusion

1. Add **noise** (**forward process**)

$x_{\text{image}} = x_0$

x_{t-1}

$q(x_t|x_{t-1})$

x_t

$\epsilon \approx x_T$

2. Learn to **denoise** it (**reverse process**)

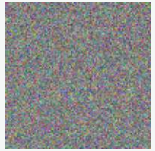
$x_{\text{image}} = x_0$

x_{t-1}

$p_\theta(x_{t-1}|x_t)$

x_t

$\epsilon = x_T$



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2. Learn to **denoise** it (**reverse process**)

$x_{\text{image}} = x_0$

We want to learn

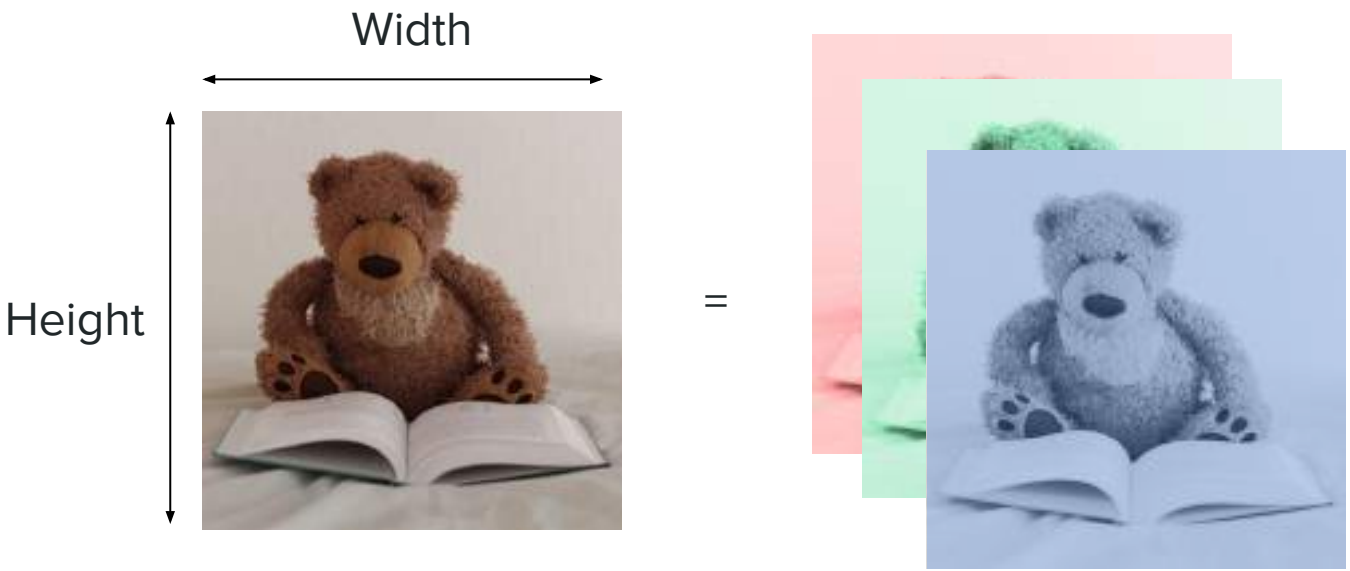
x_{t-1}

$p_\theta(x_{t-1}|x_t)$

x_t

$\epsilon = x_T$

Image representation



R **G** **B**

pixel = (0 to 255, 0 to 255, 0 to 255)

Image representation

$$\longleftrightarrow \quad x_{\text{image}} = \begin{pmatrix} x^{(1)} \\ x^{(2)} \\ \vdots \\ x^{(n)} \end{pmatrix}$$

Probability distribution in 1-D space

$$x \sim \mathcal{N}(\mu, \sigma^2)$$

Scalar

Expectation

Variance

Probability distribution in high-dim space

$$x \sim \mathcal{N}(\mu, \Sigma)$$

Vector

$$\begin{pmatrix} x^{(1)} \\ x^{(2)} \\ \vdots \\ x^{(n)} \end{pmatrix}$$

Expectation

$$\begin{pmatrix} \mu^{(1)} \\ \mu^{(2)} \\ \vdots \\ \mu^{(n)} \end{pmatrix}$$

Covariance

$$\begin{pmatrix} \Sigma^{(1,1)} & \Sigma^{(1,2)} & \dots & \Sigma^{(1,n)} \\ \Sigma^{(2,1)} & \Sigma^{(2,2)} & \dots & \Sigma^{(2,n)} \\ \vdots & \vdots & \ddots & \vdots \\ \Sigma^{(n,1)} & \Sigma^{(n,2)} & \dots & \Sigma^{(n,n)} \end{pmatrix}$$

Meaning of probability distribution in high-dim space

$$x \sim \mathcal{N}(\mu, \Sigma)$$

$$x^{(i)} \sim \mathcal{N}(\mu^{(i)}, \Sigma^{(i,i)}) \quad \text{and} \quad \mathrm{Cov}(x^{(i)}, x^{(j)}) = \Sigma^{(i,j)}$$

Good news. We will mainly (if not only) be dealing with **isotropic** Gaussians:

$$\Sigma = \sigma^2 I = \begin{pmatrix} \sigma^2 & 0 & \cdots & 0 \\ 0 & \sigma^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma^2 \end{pmatrix}$$

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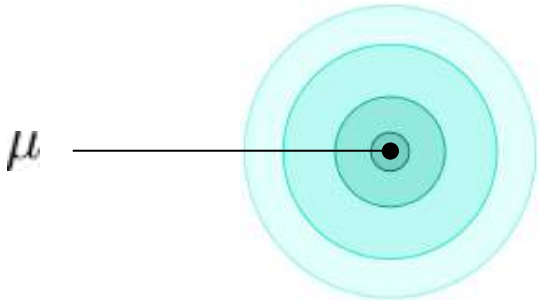

same variance
in all directions

Meaning of probability distribution in high-dim space

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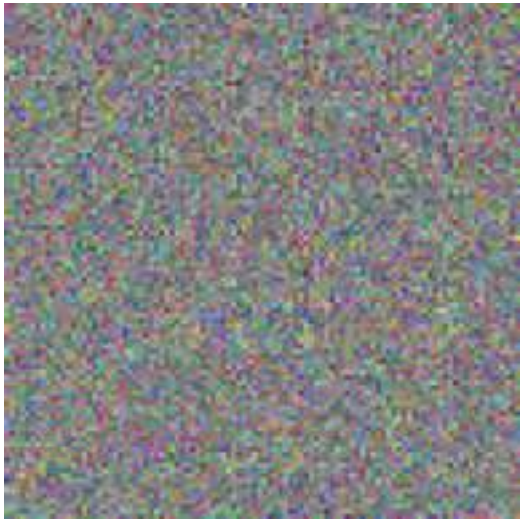
Good news. We will mainly (if not only) be dealing with **isotropic** Gaussians:

$$p(x) = \frac{1}{(2\pi\sigma^2)^{d/2}} \exp\left(-\frac{\|x-\mu\|^2}{2\sigma^2}\right)$$

probability density is known and "easy"

Noise representation



$$\longleftrightarrow \quad \epsilon = \begin{pmatrix} \epsilon^{(1)} \\ \epsilon^{(2)} \\ \vdots \\ \epsilon^{(n)} \end{pmatrix} \sim \mathcal{N}(0, I)$$

In other words, $\epsilon^{(i)} \sim \mathcal{N}(0, 1)$ independent and identically distributed

Process behind this

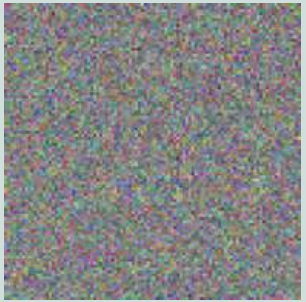
q is **chosen** by us and its **distribution** is **known** (reminder: it is stochastic!)

noisier image

$=$ something $\times$

noisy image

$+$ something else $\times$



noise

Defined by us

Process behind this

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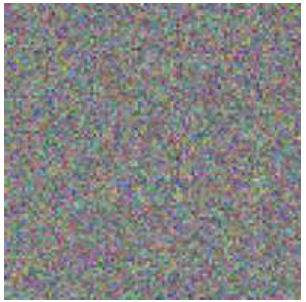


noise

"Noise schedule"

Mathematical formalism of forward process

1. Add **noise** (**forward process**)

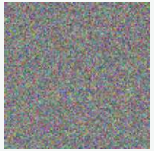
$x_{\text{image}} = x_0$

x_{t-1}

$q(x_t|x_{t-1})$

x_t

$\epsilon \approx x_T$



Noisier image

Noise

$$q(x_t|x_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t}x_{t-1}, \beta_t I)$$

Previous image

with $0 \leqslant \beta_1 < \beta_2 < \ldots < \beta_T \leqslant 1$

noise schedule

Mathematical formalism of forward process

$$q(x_t|x_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t}x_{t-1}, \beta_t I)$$

$$\downarrow$$

$$\boxed{q(x_t|x_0) = \mathcal{N}(\sqrt{\overline{\alpha}_t}x_0, (1-\overline{\alpha}_t)I)}$$

with $\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$ and $\alpha_t = 1 - \beta_t$

Derivation: Variance of sum of independent Gaussians is sum of respective variances.

Diffusion & Large Vision Models

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Variational formulation

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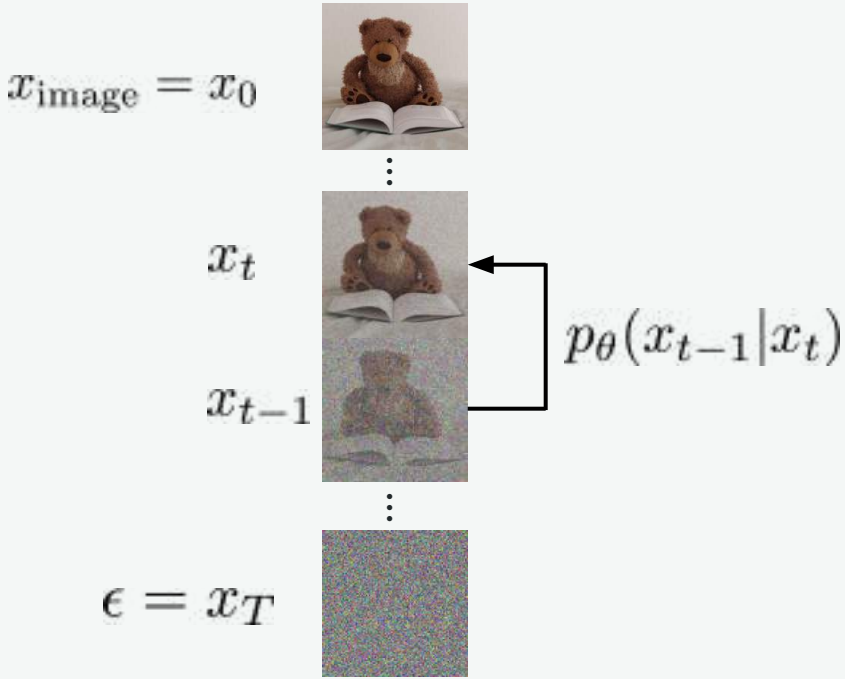
Faster sampling

Learn reverse process

2. Learn to **denoise** it (**reverse process**)



Learn reverse process

Objective. We want to learn:

$$p_\theta$$

Objective

$$\max_{\theta} \quad p_\theta(x_0)$$

find parameters θ that maximize the quantity

probability given by the model of parameters θ

training images

Objective

computationally stable +

nice properties from log

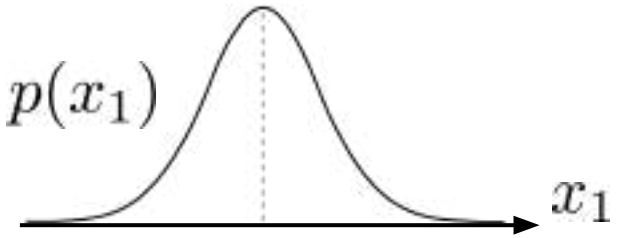
$$\max_{\theta} \log p_\theta(x_0)$$

find parameters θ that maximize the quantity

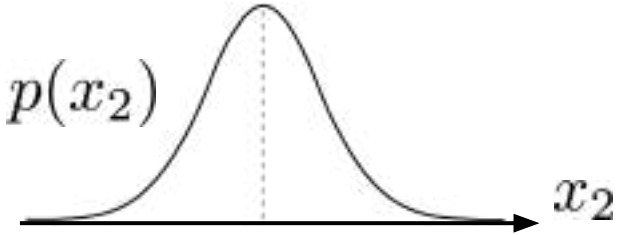
probability given by the model of parameters θ

training images

Refresher: joint probability distribution



Probability density of x_1



Probability density of x_2

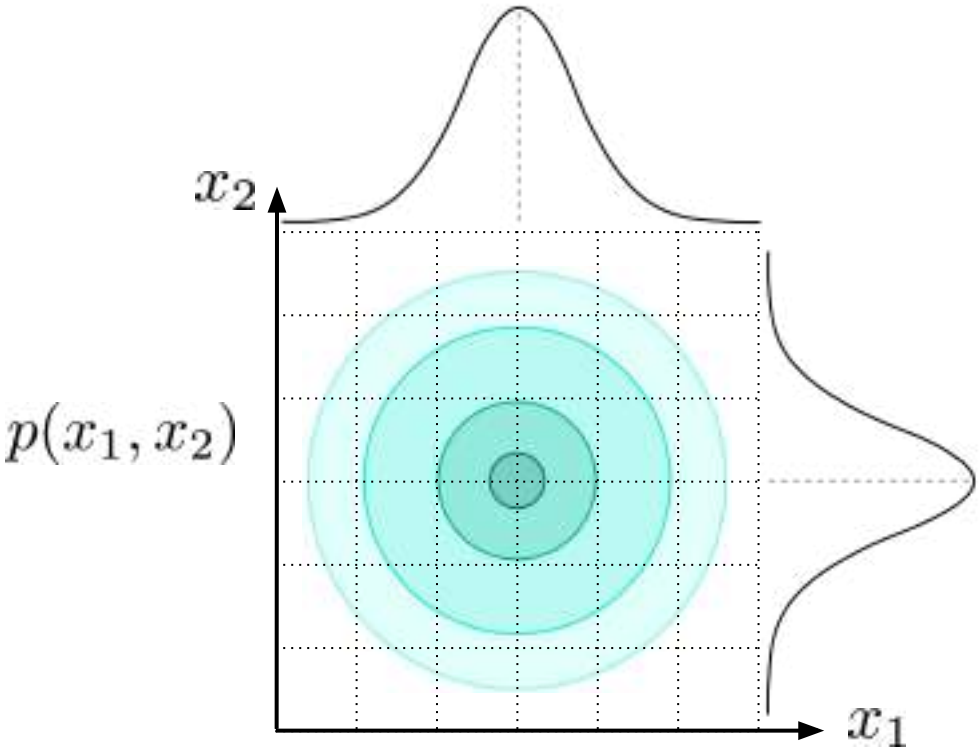
Refresher: joint probability distribution

Joint probability distribution.

$$p(x_1, x_2) = p(x_1) \times p(x_2|x_1)$$

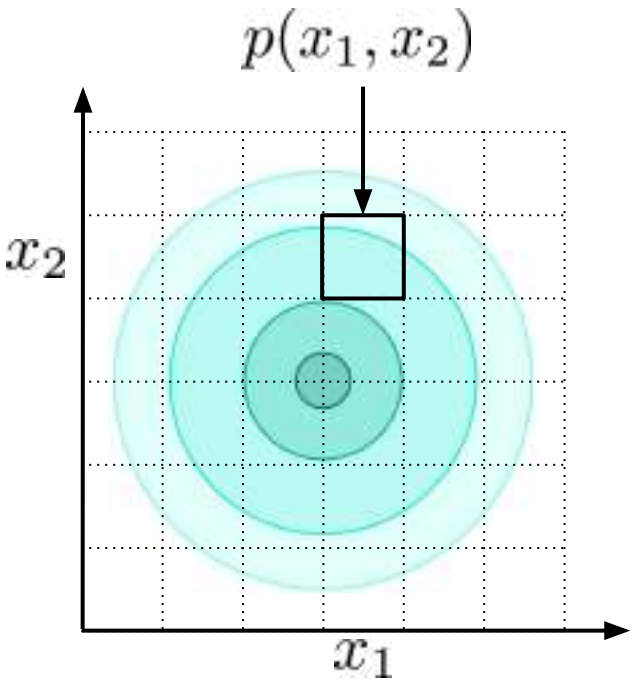
Conditional probability: if you know x_1 then what is the probability of x_2

Refresher: joint probability distribution



Joint probability $p(x_1, x_2)$ is the probability that we have x_1 **and** x_2

Refresher: joint probability distribution



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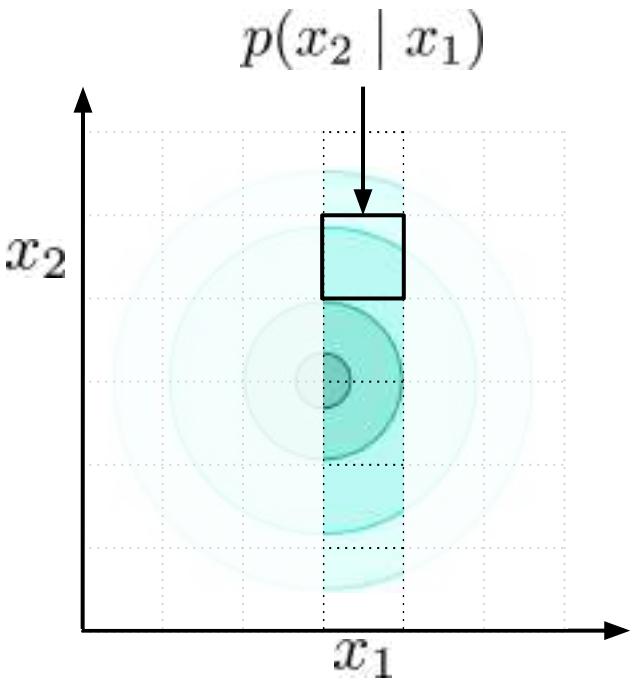


Conditional probability $p(x_2 \mid x_1)$ is the probability that we have x_2 **knowing that** we have x_1

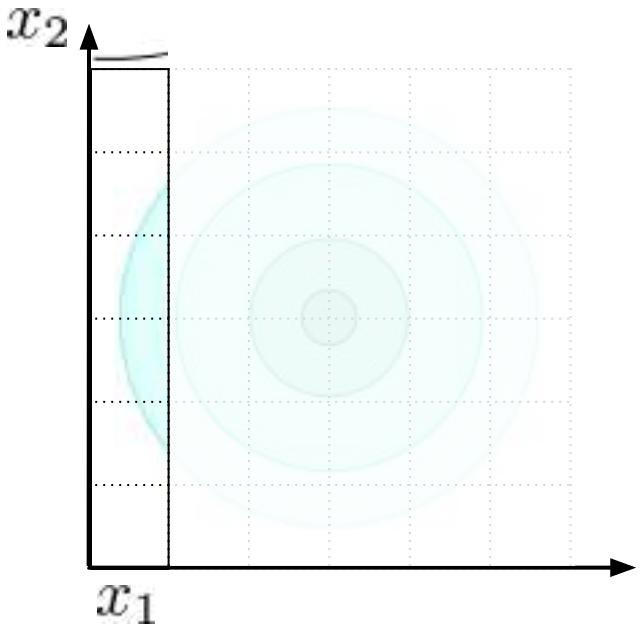
Refresher: joint probability distribution

Marginalization.

$$p(x_1) = \int p(x_1, x_2) dx_2$$

Sum across all x_2

Refresher: joint probability distribution



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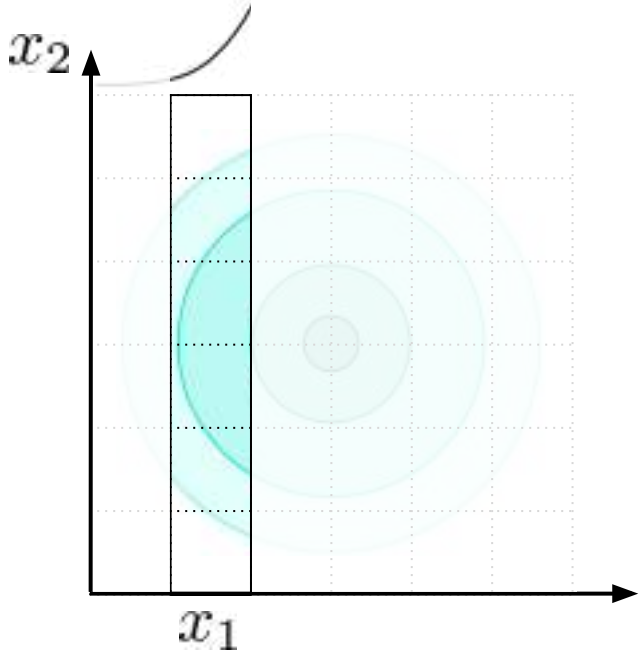

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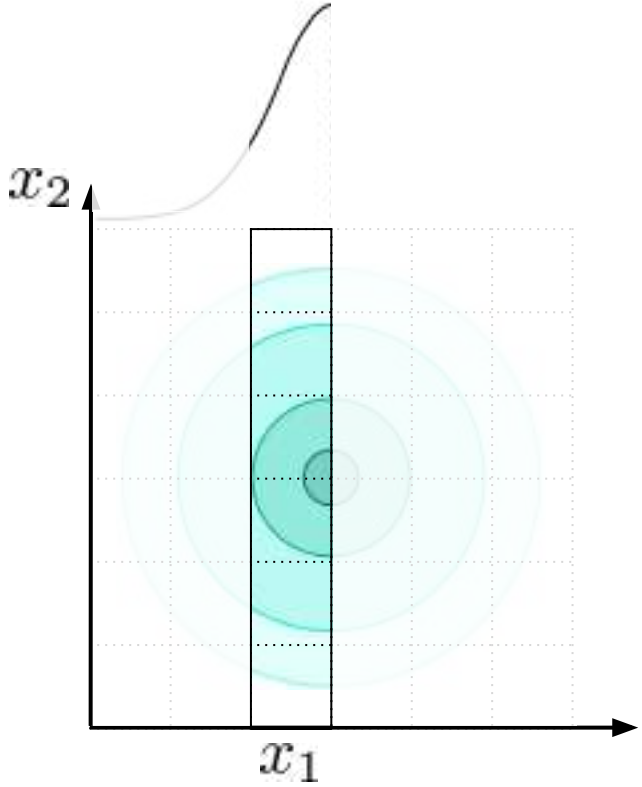

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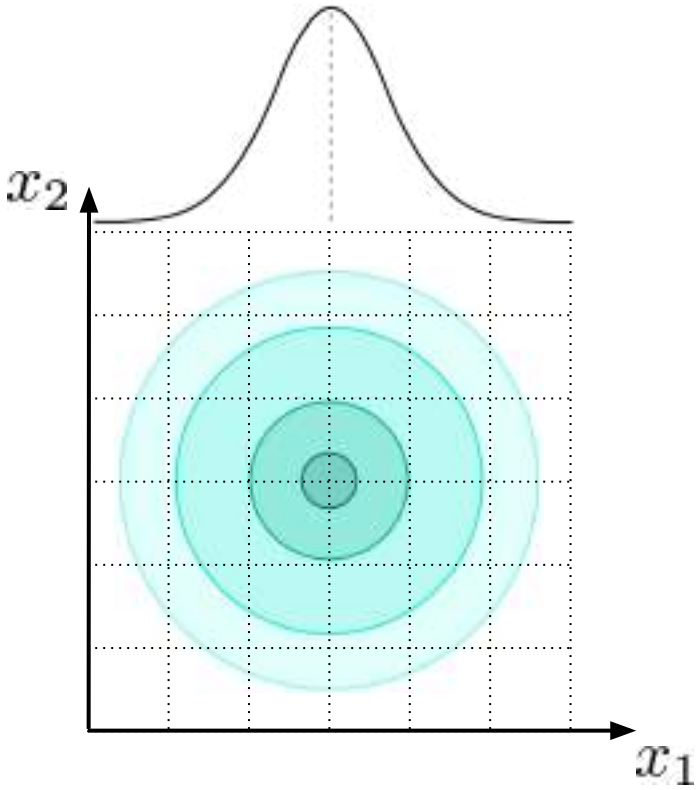

$$p(x_1) = \int p(x_1, x_2) dx_2$$

Sum across all x_2

Refresher: joint probability distribution

Joint probability distribution.

$$p(x_1, x_2, ..., x_T) = p(x_1) \times p(x_2|x_1) \times ... \times p(x_T|x_1, ..., x_{T-1})$$

Marginalization.

$$p(x_1) = \int p(x_1, x_2, ..., x_T) dx_2 ... dx_T$$

Refresher: joint probability distribution

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$$p(x_1, x_2, ..., x_T) = p(x_{1:T})$$

Marginalization.

$$p(x_1) = \int p(x_1, x_2, ..., x_T) dx_2 ... dx_T$$

Refresher: joint probability distribution

Joint probability distribution (Simplified notation).

$$p(x_{1:T}) = p(x_1) \prod_{t=2}^{T} p(x_t | x_{1:t-1})$$

Marginalization (Simplified notation).

$$p(x_1) = \int p(x_{1:T}) dx_{2:T}$$

Problem

$$\log p_\theta(x_0) = \ ?$$

Derivation of a tractable loss

Strategy.

1. Derive **lower bound** for maximum (log-)likelihood estimation
2. **Expand terms** of lower bound
3. Show lower bound is **tractable**
4. Deduce **final loss function**

Step 1: Derivation of a lower bound

1 Derive **lower bound** for maximum (log-)likelihood estimation

$$\mathbb{E}_{x_0 \sim q(x_0)}\left[\log p_\theta(x_0)\right] \boxed{\geqslant \mathbb{E}_{x_{0:T} \sim q(x_{0:T})}\left[\log \frac{p_\theta(x_{0:T})}{q(x_{1:T}|x_0)}\right]}$$

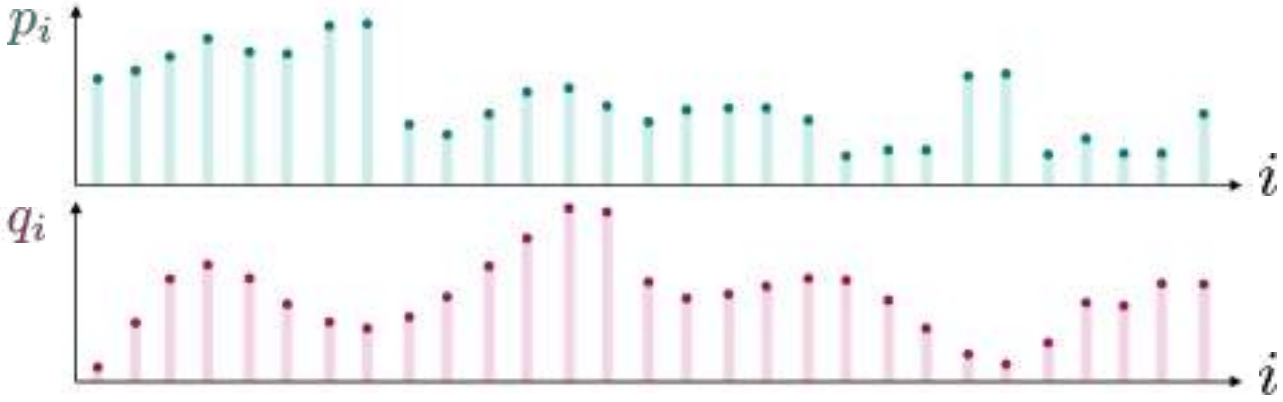
ELBO = **E**vidence **L**ower **BO**und

Derivation: Use Jensen's inequality on a convenient variational distribution

Full derivation in Denoising Diffusion Probabilistic Models, Ho et al., 2020.

Refresher: KL divergence (discrete)

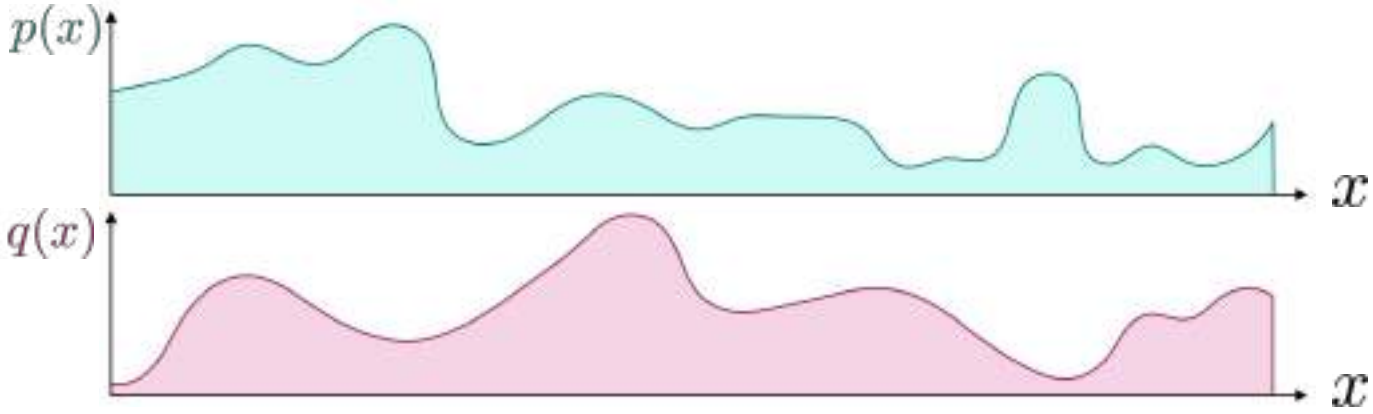
$$\mathrm{KL}(P\|Q) = \sum_{i=1}^{n} p_i \log\left(\frac{p_i}{q_i}\right)$$



Super Study Guide: Transformers & Large Language Models, Amidi, 2024.

Refresher: KL divergence (continuous)

$$\mathrm{KL}(P\|Q) = \int \log\left(\frac{p(x)}{q(x)}\right) p(x)\, dx = \mathbb{E}_{x\sim p}\left[\log\left(\frac{p(x)}{q(x)}\right)\right]$$



Super Study Guide: Transformers & Large Language Models, Amidi, 2024.

Step 2: Expand terms of the lower bound

2 **Expand terms** of lower bound

$$\text{ELBO} = -\sum_{t=2}^{T} \text{KL}\big(q(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)\big) + \text{some other terms}$$

Can we compute this?

We want to learn θ

Derivation: Use properties of the log function and rearrange terms

Full derivation in Denoising Diffusion Probabilistic Models, Ho et al., 2020.

Step 3: Show lower bound is tractable

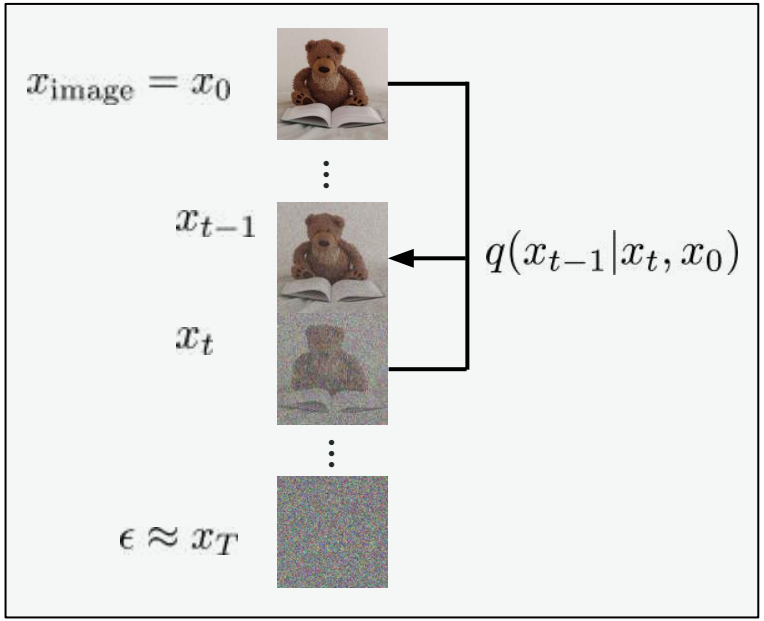
3 Show lower bound is **tractable**

a Show $q(x_{t-1} \mid x_t, x_0)$ is tractable

$$\text{ELBO} = -\sum_{t=2}^{T} \text{KL}\left(q(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)\right) + \text{some other terms}$$

Step 3: Show lower bound is tractable

3 a Show $q(x_{t-1} \mid x_t, x_0)$ is tractable



noisy image clean image

$$q(x_{t-1}|x_t, x_0)$$

less noisy image

Refresher: Bayes' rule

Likelihood

Prior

Posterior

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)}$$

Marginal likelihood, aka evidence

Step 3: Show lower bound is tractable

3 a Show $q(x_{t-1} \mid x_t, x_0)$ is tractable

$$q(x_{t-1}|x_t, x_0) = \mathcal{N}(\tilde{\mu}_t, \tilde{\beta}_t)$$

sampled noise used to go from clean to noisy image

with $\tilde{\mu}_t = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_t\right)$

noisy image

Derivation: Compute density with Bayes' rule via $q(. \mid x_0)$ + Markov property

Full derivation in Denoising Diffusion Probabilistic Models, Ho et al., 2020.

Step 3: Show lower bound is tractable

3 Show lower bound is **tractable**

a Show $q(x_{t-1} \mid x_t, x_0)$ is tractable

b Show $p_\theta(x_{t-1} \mid x_t)$ is tractable

$$\text{ELBO} = -\sum_{t=2}^{T} \text{KL}\big(q(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)\big) + \text{some other terms}$$

Step 3: Show lower bound is tractable

3 b Show $p_\theta(x_{t-1} \mid x_t)$ is tractable

We assume that p_θ is Gaussian:

$$\boxed{p_\theta(x_{t-1} \mid x_t) = \mathcal{N}\big(\mu_\theta(x_t, t), \Sigma_\theta(x_t, t)\big)}$$

Justification: If $q(x_t|x_{t-1})$ Gaussian, then $q(x_{t-1}|x_t)$ is approx. Gaussian

Full derivation in Denoising Diffusion Probabilistic Models, Ho et al., 2020.

Step 4: Deduce final loss function

4 Deduce **loss function**

$$\mathcal{L}_{\text{DDPM}} = \mathbb{E}_{t,x_0,\epsilon}\left[\left\|\epsilon_\theta\left(\sqrt{\bar{\alpha}_t}\,x_0 + \sqrt{1-\bar{\alpha}_t}\,\epsilon,\ t\right) - \epsilon\right\|^2\right]$$

$t \sim \mathcal{U}\{1, T\}$ $\quad x_0 \sim q_0(x_0)$ $\quad \epsilon \sim \mathcal{N}(0, I)$

Derivation: Compute KL divergence between two Gaussians

Full derivation in Denoising Diffusion Probabilistic Models, Ho et al., 2020.

Summary of what we have done

Strategy.

1. Derive **lower bound** for maximum (log-)likelihood estimation ← **ELBO!**
2. **Expand terms** of lower bound ← **Surfaced KL divergence**
3. Show lower bound is **tractable** ← **Bayes' rule + Gaussian properties**
4. Deduce **final loss function** ← **Incredibly simple noise prediction!**

Diffusion & Large Vision Models

Motivation

Forward / backwards process

Variational formulation

Training

Inference

Faster sampling

Training recipe

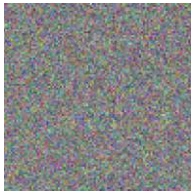
1. Sample:

clean image

$x_0 \sim q_0(x_0)$

noise

$\epsilon \sim \mathcal{N}(0, I)$



time step

$t \sim \mathcal{U}\{1, T\}$

noised image

$$x_t = \sqrt{\overline{\alpha}_t} x_0 + \sqrt{1 - \overline{\alpha}_t} \epsilon$$

Training recipe

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clean image

$x_0 \sim q_0(x_0)$

noise

$\epsilon \sim \mathcal{N}(0, I)$

time step

$t \sim \mathcal{U}\{1, T\}$

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$$x_t = \sqrt{\overline{\alpha}_t} x_0 + \sqrt{1 - \overline{\alpha}_t} \epsilon$$

2. Use x_t and t to **predict** ϵ via $\epsilon_\theta(x_t, t)$

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clean image

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time step

$t \sim \mathcal{U}\{1, T\}$

noised image

$x_t = \sqrt{\overline{\alpha}_t} x_0 + \sqrt{1 - \overline{\alpha}_t} \epsilon$

2. Use x_t and t to **predict** ϵ via $\epsilon_\theta(x_t, t)$

3. Compute **loss** $\mathcal{L} = ||\epsilon_\theta(x_t, t) - \epsilon||^2$ and **backpropagate** through ϵ_θ

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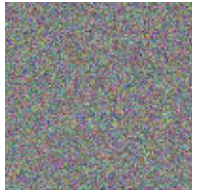
Inference recipe

1. Sample:

noise

$x_T \sim \mathcal{N}(0, I)$



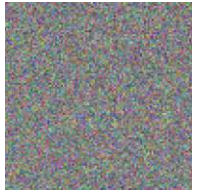
2. Perform **iterative update** from x_t to x_{t-1}

Inference recipe

1. Sample:

noise

$x_T \sim \mathcal{N}(0, I)$



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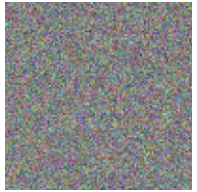

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x_t

$\epsilon_\theta(x_t, t)$

Predicted noise
to remove

Inference recipe

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noise

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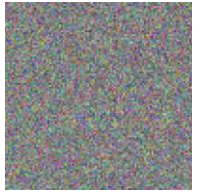


2. Perform **iterative update** from x_t to x_{t-1}

$$x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, t)$$

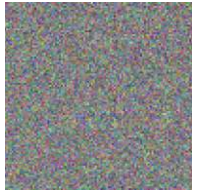
Predicted noise
to remove

Inference recipe

1. Sample:

noise

$x_T \sim \mathcal{N}(0, I)$



2. Perform **iterative update** from x_t to x_{t-1}

$$\frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, t)\right) + \sigma_t z$$

Predicted noise
to remove

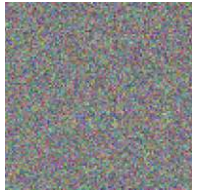
Sampled from a
Normal distribution

Inference recipe

1. Sample:

noise

$x_T \sim \mathcal{N}(0, I)$



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noise

$$x_T \sim \mathcal{N}(0, I)$$

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$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, t)\right) + \sigma_t z$$

3. Obtain **final** image x_0

Landmark paper that is a must-read

DDPM = **D**enoising **D**iffusion **P**robabilistic **M**odels

Denoising Diffusion Probabilistic Models

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Pieter Abbeel
UC Berkeley
pabbeel@cs.berkeley.edu

Abstract

We present high quality image synthesis results using diffusion probabilistic models, a class of latent variable models inspired by considerations from nonequilibrium thermodynamics. Our best results are obtained by training on a weighted variational bound designed according to a novel connection between diffusion probabilistic models and denoising score matching with Langevin dynamics, and our models naturally admit a progressive lossy decompression scheme that can be interpreted as a generalization of autoregressive decoding. On the unconditional CIFAR10 dataset, we obtain an Inception score of 9.46 and a state-of-the-art FID score of 3.17. On 256x256 LSUN, we obtain sample quality similar to ProgressiveGAN. Our implementation is available at https://github.com/hojonathanho/diffusion.

Denoising Diffusion Probabilistic Models, Ho et al., 2020.

Diffusion & Large Vision Models

Motivation

Forward / backwards process

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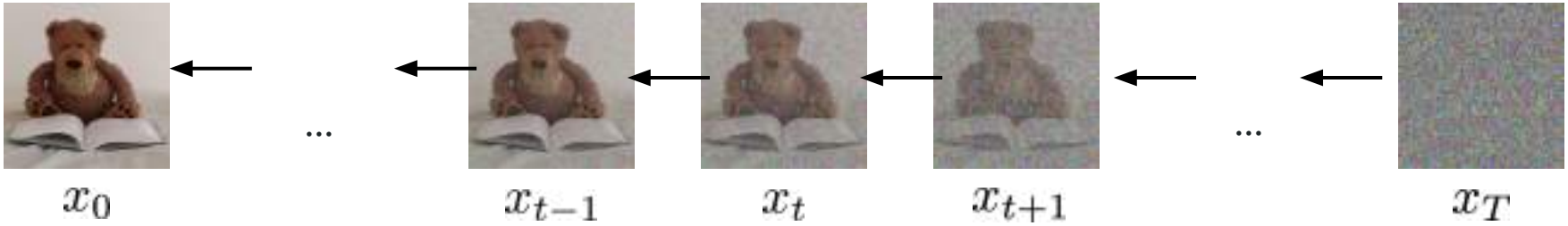
Faster sampling

Limitations

Generation process $p_\theta(x_{t-1} \mid x_t)$ $T \gg 1$



hojonathanho/diffusion, official GitHub repository of the DDPM paper, 2020.

Limitations

Orders of magnitude

- $O(T)$ more expensive than VAE/GAN
- $T = 1000$ → possibly spend ~minutes per sample

hojonathanho/diffusion, official GitHub repository of the DDPM paper, 2020.

Limitations

Generation process $p_\theta(x_{t-1} \mid x_t)$ $T \gg 1$

Orders of magnitude

- $O(T)$ more expensive than VAE/GAN
- $T = 1000$ → possibly spend ~minutes per sample

In practice, DDPM is too slow at inference time!

Mitigation attempts

Starting point. Recall iterative generative formula between two steps:

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, t)\right) + \sigma_t z$$

Predicted noise to remove

Sampled from a Normal distribution

Mitigation attempts

Attempt. What if we derive a formula by induction?

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, t)\right) + \sigma_t z$$

$$x_{t-2} = \frac{1}{\sqrt{\alpha_{t-1}}}\left(x_{t-1} - \frac{1-\alpha_{t-1}}{\sqrt{1-\bar{\alpha}_{t-1}}}\epsilon_\theta(x_{t-1}, t-1)\right) + \sigma_{t-1} z$$

$$\vdots$$

$$x_{\tau_{i-1}} = A x_{\tau_i} + \sum_{s=\tau_{i-1}+1}^{\tau_i} B_s \epsilon_\theta(x_s, s) + C_s z_s$$

Mitigation attempts

Attempt. What if we derive a formula by induction?

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, t)\right) + \sigma_t z$$

$$x_{t-2} = \frac{1}{\sqrt{\alpha_{t-1}}}\left(x_{t-1} - \frac{1-\alpha_{t-1}}{\sqrt{1-\bar{\alpha}_{t-1}}}\epsilon_\theta(x_{t-1}, t-1)\right) + \sigma_{t-1} z$$

$$\vdots$$

$$x_{\tau_{i-1}} = Ax_{\tau_i} + \sum_{s=\tau_{i-1}+1}^{\tau_i} B_s \boxed{\epsilon_\theta(x_s, s)} + C_s z_s$$

Problem: Still need to evaluate function several times. No progress.

Mitigation attempts

Attempt 2. What if we directly skip steps?

$$\tau_0 = 0 < \tau_1 < \ldots < \tau_{S-1} < \tau_S = T$$

$$x_{\tau_{i-1}} = \sqrt{\frac{\bar{\alpha}_{\tau_{i-1}}}{\bar{\alpha}_{\tau_i}}} \left(x_{\tau_i} - \frac{1 - \bar{\alpha}_{\tau_i} / \bar{\alpha}_{\tau_{i-1}}}{\sqrt{1 - \bar{\alpha}_{\tau_i}}} \epsilon_\theta(x_{\tau_i}, \tau_i) \right) + \sigma_{\tau_i} z$$

Denoising Diffusion Implicit Models, Song et al., 2020.

Mitigation attempts

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Problem: mix large jumps + stochasticity = bad quality

Problem reformulation

Follow-up idea. How to remove stochasticity while making these larger jumps?

Key property. Loss function only relies on marginals $q(x_t \mid x_0)$

Problem statement. Find q such that:

- marginal is the same as in DDPM
- generation process can be made deterministic

Problem reformulation

$$q_\sigma(x_t \mid x_0) = q(x_t \mid x_0)$$

$$q_\sigma(x_{t-1} \mid x_t, x_0)$$

$$p_\theta(x_{t-1} \mid x_t)$$ can be made deterministic

$$q_\sigma(x_{t-1} \mid x_t, x_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_{t-1}}\, x_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2}\, \frac{x_t - \sqrt{\bar{\alpha}_t}\, x_0}{\sqrt{1 - \bar{\alpha}_t}},\ \sigma_t^2 I\right)$$

Problem reformulation

$$q_\sigma(x_{t-1} \mid x_t, x_0)$$

$$q_\sigma(x_t \mid x_0) = q(x_t \mid x_0)$$

Justification: proof by induction

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Problem reformulation

$$q_\sigma(x_t \mid x_0) = q(x_t \mid x_0)$$

Justification: proof by induction

$$q_\sigma(x_{t-1} \mid x_t, x_0)$$

$p_\theta(x_{t-1} \mid x_t)$ can be made deterministic

$$p_\theta(x_{t-1} \mid x_t) \triangleq q_{\sigma=0}(x_{t-1} \mid x_t, \hat{x}_0(x_t))$$

$$q_\sigma(x_{t-1} \mid x_t, x_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_{t-1}}\, x_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \cancel{\sigma_t^2}}\, \frac{x_t - \sqrt{\bar{\alpha}_t}\, x_0}{\sqrt{1 - \bar{\alpha}_t}},\ \cancel{\sigma_t^2} I\right)$$

Why is DDIM called that way?

DDIM = **D**enoising **D**iffusion **I**mplicit **M**odels

$$\sigma_t = 0$$

$$\hat{x}_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\,\epsilon_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}}$$

"our prediction of the clean image from time t"

Derivation: use the marginal distribution formula

Why is DDIM called that way?

DDIM = **D**enoising **D**iffusion **I**mplicit **M**odels

$$\sigma_t = 0$$

$$\hat{x}_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\,\epsilon_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}}$$

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\,\hat{x}_0 + \sqrt{1-\bar{\alpha}_{t-1}}\,\epsilon_\theta(x_t, t) \quad + \quad \varnothing$$

"Our prediction of the clean image from time t"

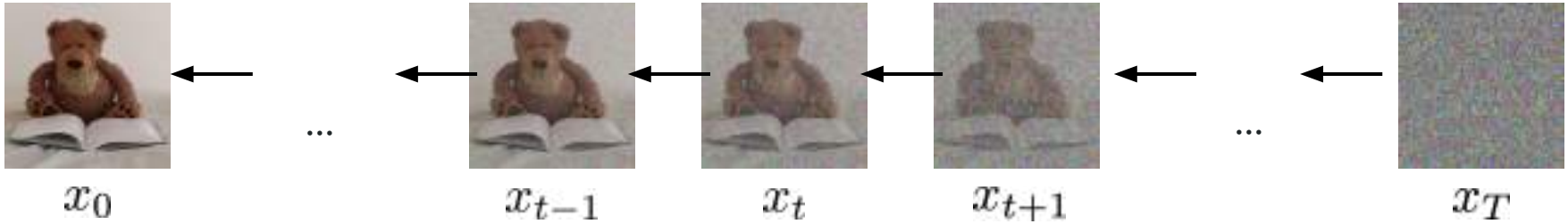
"...that we noise to time t-1..."

... with no stochasticity!

Why is the DDIM formulation useful?

Update rule

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\,\hat{x}_0(t) + \sqrt{1-\bar{\alpha}_{t-1}}\,\epsilon_\theta(x_t, t)$$



Why is the DDIM formulation useful?

Update rule: accelerated!

$$x_{\tau_{i-1}} = \sqrt{\bar{\alpha}_{\tau_{i-1}}}\,\hat{x}_0(\tau_i) + \sqrt{1-\bar{\alpha}_{\tau_{i-1}}}\,\epsilon_\theta(x_{\tau_i}, \tau_i)$$

$$\tau_0 = 0 < \tau_1 < \ldots < \tau_{S-1} < \tau_S = T$$

$S \ll T$ with $\frac{T}{S} \triangleq$ "speedup"

Applications

Speedup

- Typically 10x-100x
- Need to trade off speed with quality

ermongroup/ddim, official GitHub repository of the DDIM paper, 2020.

Applications

Speedup

- Typically 10x-100x
- Need to trade off speed with quality

Quality (based on CIFAR10 experiments)

Speed-up	1x	10x	20x	50x	100x
FID impact	baseline	+3%	+16%	+70%	+330%

Denoising Diffusion Implicit Models, Song et al., 2020.

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Discussion

- Skipping steps with inter-steps stochasticity → much worse quality
- Schedule for $\{\tau_i\}$ can be chosen as a hyperparameter

Denoising Diffusion Implicit Models, Song et al., 2020.

DDIM conclusion

Recipe

1. Choose "convenient" modeling assumptions compatible with DDPM optimization
2. Remove inter-step stochasticity = DDIM
3. Skip steps!

$$\boxed{p_\theta(x_{t-1} \mid x_t) \longrightarrow p_\theta(x_{\tau_{i-1}} \mid x_{\tau_i})}$$

Thank you for your attention!